

Shubhadeep Mandal

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EDUCATION

International Institute of Information Technology, Hyderabad <i>Telangana, India</i> <i>Master of Technology in Computer Science and Engineering</i>	July 2025 – Present <i>CGPA: 9.12</i>
Indian Institute of Information Technology, Ranchi <i>Ranchi, India</i> <i>Bachelor of Technology (Hons.) in Computer Science and Engineering</i>	Aug 2020 – Jun 2024 <i>CGPA: 9.05</i>

PROJECTS

- AmbiStory: NLP Plausibility & Disambiguation Engine** | *Python, PyTorch, Transformers, NLI* 🔄
- Achieved the best non-LLM solution on the AmbiStory problem (SemEval 2026), reaching $\rho = 0.535$ and 72.2% test accuracy.
 - Engineered a 202-dim vector fusing multi-context sentence embeddings, NLI priors, and PCA interactions into an ordinal regression head.
 - Mitigated selection bias ($+0.265 \rho$) via homonym-grouped 5-fold CV, annotator distribution KL loss, and Stochastic Weight Averaging (SWA).
- Football Video Analytics with YOLO** | *Python, YOLOv8, OpenCV, Optical Flow, K-Means* 🔄
- Fine-tuned YOLOv8 for multi-object tracking of players, ball, and referees, with K-Means color clustering for team assignment.
 - Compensated camera pan/tilt using Lucas-Kanade optical flow and mapped pixel coordinates to pitch meters via homography.
 - Engineered continuous speed, distance covered, and team possession telemetry pipelines with real-time video overlay generation.
- WatchTower AI: Real-Time Edge Object Detection** | *Next.js 14, TypeScript, TensorFlow.js, WebRTC* 🔄
- Built an in-browser edge object detection web application using TensorFlow.js and quantized COCO-SSD, achieving 30 FPS zero-latency inference.
 - Processed live camera feeds via WebRTC APIs, rendering dynamic bounding boxes, confidence scoring, and snapshot recording on HTML5 Canvas.
- Peer-to-Peer Distributed File Sharing System** | *C++, Sockets, Multithreading, OpenSSL* 🔄
- Built a BitTorrent-style P2P system in C++, achieving an 8.5x speedup on 1GB transfers (193s to 22.8s) across 5 parallel peers.
 - Engineered out-of-order parallel chunk writing via `pwrite()` and dynamic bitfield generation for simultaneous partial seeding.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, SQL, Bash.
NLP & Deep Learning: Transformers, PyTorch, Hugging Face, Sentence-Transformers, NLI, Ordinal Regression.
Computer Vision & Edge ML: YOLOv8, OpenCV, TensorFlow.js, Optical Flow, K-Means, Streamlit.
Systems, Data & Tools: POSIX Sockets, Linux/Unix, Git, NumPy, Pandas, Scikit-learn, Matplotlib.

ACHIEVEMENTS

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

INTERESTS

- Web Novel and Manga Enthusiast.
- Chess and Badminton.